

Q1: Select the most correct answer: (15 points)

1. What's the equivalent of $t2 = A[A[0]]$ in assembly where the base address of A is in $s0$:

a. lw $t0, 0(s0)$
lw $t2, 0(t0)$

b. lw $t0, 0(s0)$ $\rightarrow t0 \rightarrow A[0]$ $\rightarrow A[A[0]]$
sll $t1, t0, 2$ $\rightarrow t1 = t0 \times 4$
add $t1, s0, t1$ $\rightarrow t1 = A[t0 \times 4]$
lw $t2, 0(t1)$ $\rightarrow t2 = A[t0 \times 4]$

c. lw $t0, 0(s0)$
add $t1, s0, t0$
lw $t2, 0(t1)$

d. lw $t2, 0(s0)$

21/30

2. Which of the following instructions uses PC-relative addressing?

a. addi $t1, t2, 10$
c. jr sra

b. lw $t0, 4(s0)$
d. beq $t0, t1, LABEL$

3. Assume a new instruction **move $t0, t1$** that copies the contents of $t1$ to $t0$. Which of the following standard MIPS instructions is **NOT** equivalent to this operation?

a. add $t0, t1, zero$ ✓
c. sub $t0, t1, zero$ $\rightarrow t0 = t1 - 0 = t1$

b. addi $t0, t1, 0$ ✓
d. sub $t0, zero, t1$ $\rightarrow t0 = 0 - t1 = -t1$

4. Decode the machine instruction **0x014B4820**:

a. add $t1, t2, t3$
c. and $t1, t2, t2$

b. addi $t1, t2, 20$
d. sub $t1, t2, t3$

5. Which of the following registers is used with one of J-type instructions?

a. $t0$

c. $s0$

b. sra

d. $s3$

0000 0000

6. What's the final value stored in $t0$ after executing **lui $t0, 0x1234$** , assuming $t0$ originally contains 0?

a. 0x12340000

b. 0x00001234

c. 0xFFFF1234

d. 0x1234FFFF

PC + 4

7. If a **beq $t0, t1, Label$** at address 0x00400020 branches to 0x00400028, what is the value of **Label**?

a. 1

b. 6

c. 4

d. 8

Stu

8. In which case can signed overflow occur in the operation $A - B$:

- a. A is positive and B is positive
- b. A = 0 and B = 0
- c. A is positive and B is negative
- d. A is negative and B is negative

$P - N$
 $N - N$
 $N + N$
 $N + N$
 $0 - 0$

large
small

9. Assume a 4-bits multiplicand and a 4-bits multiplier, what's the size of the ALU that could perform the first-version multiplication :

- a. 8-bits ALU
- b. 16-bits ALU
- c. 4-bits ALU
- d. 32-bits ALU

11 8 23

10. What is the IEEE 754 single-precision representation of the decimal value -1.5 ?

- a. 1011 1111 0110 0000 0000 0000 0000 0000
- b. 0100 0000 0100 0000 0000 0000 0000 0000
- c. 1000 0000 0110 0000 0000 0000 0000 0000
- d. 1011 1111 1100 0000 0000 0000 0000 0000

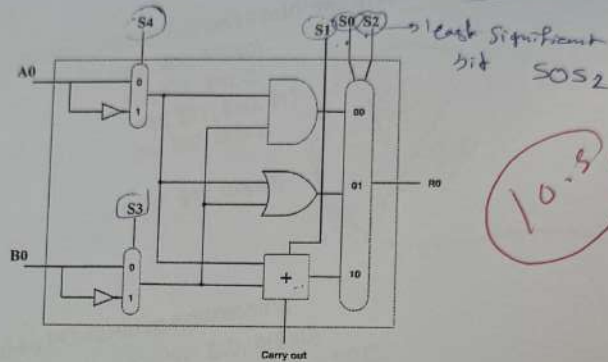
-1.1
 2^0
 2^1
 124
 126
 0.5
 1

Write your choice in the following Answers table. Other answers located outside the table will not be considered.

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	d	d	a	b	a	c	A	A	d

10.5

Q2: Based on Chapter 3 (ALU Design). Your goal is to understand the following design and answer the questions associated to it. Consider the distribution of the selectors (15 points)



1. [2.5 points] Write down input signals to perform subtraction ($A - B$):

S4	S3	S2	S1	S0
0	1	1	1	0

1.5

2. [3.5 points] Write down input signals to support a new instruction called "INC A" which increments A by 1 ($A = A + 1$):

A0	A1	A2	A3	S4	S3	S2	S1	S0
0	0	0	0	0	0	0	1	1

3.5

3. [3.5 points] Write down input signals to support a new instruction called "DEC A" which decrements A by 1 ($A = A - 1$):

A0	A1	A2	A3	S4	S3	S2	S1	S0
1	1	1	1	0	1	0	1	1

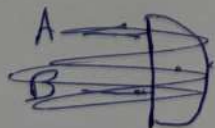
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4. [3.5 points] Write down input signals to support a new instruction called "INV A" which inverts A. For example, if $A = 0001$, then INV A returns 1110:

A0	A1	A2	A3	S4	S3	S2	S1	S0
0	0	0	1	1	0	0	1	0

2.5

5. [2 points] Draw the logic circuit diagram for the carryout in the full adder:



$$\text{Carryin} = A \cdot B + B \cdot \text{carryin}$$

A	B	Carryin	Carryout
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	1

A	B	Carryin	Carryout
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	1